Moreover, the numbers

$$L(\beta, \alpha, \gamma), \qquad L(\alpha, \gamma, \beta), \qquad \text{and} \qquad L(\gamma, \beta, \alpha)$$

are all equal to $-L(\alpha, \beta, \gamma)$. Therefore, L vanishes for all possible configurations of these three points. Q.E.D.

Exercises for Section 7.2

1. Let $f : [a, b] \to \mathbb{R}$. Show that $f \notin \mathcal{R}[a, b]$ if and only if there exists $\varepsilon_0 > 0$ such that for every $n \in \mathbb{N}$ there exist tagged partitions $\dot{\mathcal{P}}_n$ and $\dot{\mathcal{Q}}_n$ with $||\dot{\mathcal{P}}_n|| < 1/n$ and $||\dot{\mathcal{Q}}_n|| < 1/n$ such that $\left|S(f; \dot{\mathcal{P}}_n) - S(f; \dot{\mathcal{Q}}_n)\right| \geq \varepsilon_0$.
2. Consider the function h defined by $h(x) := x + 1$ for $x \in [0, 1]$ rational, and $h(x) := 0$ for $x \in [0, 1]$ irrational. Show that h is not Riemann integrable.
3. Let $H(x) := k$ for $x = 1/k$ ($k \in \mathbb{N}$) and $H(x) := 0$ elsewhere on [0, 1]. Use Exercise 1, or the argument in 7.2.2(b), to show that H is not Riemann integrable.
4. If $\alpha(x) := -x$ and $\omega(x) := x$ and if $\alpha(x) \leq f(x) \leq \omega(x)$ for all $x \in [0, 1]$, does it follow from the Squeeze Theorem 7.2.3 that $f \in \mathcal{R}[0, 1]$?
5. If J is any subinterval of $[a, b]$ and if $\varphi_J(x) := 1$ for $x \in J$ and $\varphi_J(x) := 0$ elsewhere on $[a, b]$, we say that φ_J is an *elementary step function* on $[a, b]$. Show that every step function is a linear combination of elementary step functions.
6. If $\psi : [a, b] \to \mathbb{R}$ takes on only a finite number of distinct values, is ψ a step function?
7. If $S(f; \dot{\mathcal{P}})$ is any Riemann sum of $f : [a, b] \to \mathbb{R}$, show that there exists a step function $\varphi : [a, b] \to \mathbb{R}$ such that $\int_a^b \varphi = S(f; \dot{\mathcal{P}})$.
8. Suppose that f is continuous on $[a, b]$, that $f(x) \geq 0$ for all $x \in [a, b]$ and that $\int_a^b f = 0$. Prove that $f(x) = 0$ for all $x \in [a, b]$.
9. Show that the continuity hypothesis in the preceding exercise cannot be dropped.
10. If f and g are continuous on $[a, b]$ and if $\int_a^b f = \int_a^b g$, prove that there exists $c \in [a, b]$ such that $f(c) = g(c)$.
11. If f is bounded by M on $[a, b]$ and if the restriction of f to every interval $[c, b]$ where $c \in (a, b)$ is Riemann integrable, show that $f \in \mathcal{R}[a, b]$ and that $\int_c^b f \to \int_a^b f$ as $c \to a+$. [*Hint:* Let $\alpha_c(x) := -M$ and $\omega_c(x) := M$ for $x \in [a, c)$ and $\alpha_c(x) := \omega_c(x) := f(x)$ for $x \in [c, b]$. Apply the Squeeze Theorem 7.2.3 for c sufficiently near a.]
12. Show that $g(x) := \sin(1/x)$ for $x \in (0, 1]$ and $g(0) := 0$ belongs to $\mathcal{R}[0, 1]$.
13. Give an example of a function $f : [a, b] \to \mathbb{R}$ that is in $\mathcal{R}[c, b]$ for every $c \in (a, b)$ but which is not in $\mathcal{R}[a, b]$.
14. Suppose that $f : [a, b] \to \mathbb{R}$, that $a = c_0 < c_1 < \cdots < c_m = b$ and that the restrictions of f to $[c_{i-1}, c_i]$ belong to $\mathcal{R}[c_{i-1}, c_i]$ for $i = 1, \ldots, m$. Prove that $f \in \mathcal{R}[a, b]$ and that the formula in Corollary 7.2.11 holds.
15. If f is bounded and there is a finite set E such that f is continuous at every point of $[a, b] \backslash E$, show that $f \in \mathcal{R}[a, b]$.
16. If f is continuous on $[a, b]$, $a < b$, show that there exists $c \in [a, b]$ such that we have $\int_a^b f = f(c)(b - a)$. This result is sometimes called the *Mean Value Theorem for Integrals*.
17. If f and g are continuous on $[a, b]$ and $g(x) > 0$ for all $x \in [a, b]$, show that there exists $c \in [a, b]$ such that $\int_a^b fg = f(c) \int_a^b g$. Show that this conclusion fails if we do not have $g(x) > 0$. (Note that this result is an extension of the preceding exercise.)

18. Let f be continuous on $[a, b]$, let $f(x) \geq 0$ for $x \in [a, b]$, and let $M_n := (\int_a^b f^n)^{1/n}$. Show that $\lim(M_n) = \sup\{f(x) : x \in [a, b]\}$.
19. Suppose that $a > 0$ and that $f \in \mathcal{R}[-a, a]$.
    (a) If f is *even* (that is, if $f(-x) = f(x)$ for all $x \in [0, a]$), show that $\int_{-a}^{a} f = 2\int_0^a f$.
    (b) If f is *odd* (that is, if $f(-x) = -f(x)$ for all $x \in [0, a]$), show that $\int_{-a}^{a} f = 0$.
20. If f is continuous on $[-a, a]$, show that $\int_{-a}^{a} f(x^2)dx = 2\int_0^a f(x^2)dx$.

Section 7.3 The Fundamental Theorem

We will now explore the connection between the notions of the derivative and the integral. In fact, there are *two* theorems relating to this problem: one has to do with integrating a derivative, and the other with differentiating an integral. These theorems, taken together, are called the Fundamental Theorem of Calculus. Roughly stated, they imply that the operations of differentiation and integration are inverse to each other. However, there are some subtleties that should not be overlooked.

The Fundamental Theorem (First Form)

The First Form of the Fundamental Theorem provides a theoretical basis for the method of calculating an integral that the reader learned in calculus. It asserts that if a function f is the derivative of a function F, *and if* f *belongs to* $\mathcal{R}[a, b]$, then the integral $\int_a^b f$ can be calculated by means of the evaluation $F|_a^b := F(b) - F(a)$. A function F such that $F'(x) = f(x)$ for all $x \in [a, b]$ is called an **antiderivative** or a **primitive of** f on $[a, b]$. Thus, when f has an antiderivative, it is a very simple matter to calculate its integral.

In practice, it is convenient to allow some exceptional points c where $F'(c)$ does not exist in $\mathbb{R}$, or where it does not equal $f(c)$. It turns out that we can permit a *finite* number of such exceptional points.

7.3.1 Fundamental Theorem of Calculus (First Form) *Suppose there is a* ***finite*** *set* $\boldsymbol{E}$ *in* $[a, b]$ *and functions* f, $F := [a, b] \to \mathbb{R}$ *such that*:

(a) F *is continuous on* $[a, b]$,

(b) $F'(x) = f(x)$ *for all* $x \in [a, b] \backslash E$,

(c) f *belongs to* $\mathcal{R}[a, b]$.

Then we have

$$\int_a^b f = F(b) - F(a). \tag{1}$$

Proof. We will prove the theorem in the case where $E := \{a, b\}$. The general case can be obtained by breaking the interval into the union of a finite number of intervals (see Exercise 1).

Let $\varepsilon > 0$ be given. Since $f \in \mathcal{R}[a, b]$ by assumption (c), there exists $\delta_\varepsilon > 0$ such that if $\dot{\mathcal{P}}$ is *any* tagged partition with $||\dot{\mathcal{P}}|| < \delta_\varepsilon$, then

$$\left| S(f; \dot{\mathcal{P}}) - \int_a^b f \right| < \varepsilon. \tag{2}$$